

Table 26: The comparisons of agent results based on different LLMs.

		LLaMA2-7B	LLaMA2 70B	GPT-3.5-turbo + LangChain (ours)	GPT-4
Configuration	Module Selection	Possible operations: Aggregation: . . . , Pooling: . . . , Readout: . . .	Here are my suggestions for the three operations provided, along with their justifications, for the graph-level task: Convolution: . . . , Pooling: . . . , Readout: . . .	The response list is ['aggregation', 'pooling', 'readout', 'selection', 'fusion']	The graph-level task involves learning a representation for the entire graph, which is often used for graph classification or regression. Convolution: Yes. . . . , Pooling: Yes. . . . , Readout: Yes.So, all the operations [convolution, pooling, readout] can be used for graph-level tasks.
	Operation Preparation & Candidate Selection	To address your request: 1. Justification: . . . , 2. Finding the corresponding class: . . . , 3. Returning the class name and module name	The class name for the convolution operation would be ChebConv. The module name would be pyg.nn.	'ChebConv', 'torch_geometric.nn.conv.cheb_conv.ChebConv'	This function is part of the convolution module in PyTorch Geometric. So, the class name is ChebConv and the module name is convolution.
	Formatted output	{ 'aggregation': ['GCNConv', 'GATConv', 'ChebConv'], 'READOUT': . . . }	{ 'aggregation': ['GCNConv', 'GATConv', 'ChebConv'], 'READOUT': . . . }	{ 'aggregation': [GCN, SAGE, ChebConv], 'readout': . . . }	{ 'aggregation': ['GCNConv', 'GATConv', 'ChebConv'], 'READOUT': . . . }
Algorithm	Identifying Requirements & Algorithm Selection	Firstly, let me clarify the difference between them. . . . Then, Lets evaluate the options based on these principles. . . .	Recommendation: Differentiable Search Algorithm	You should use "differentiable search algorithm".	. . . So, in this case, both differentiable and random search algorithms should be used depending on the specific operation.